

Session 3.4

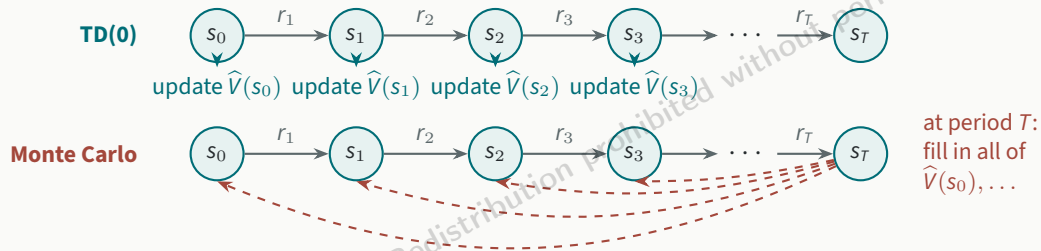
Temporal Difference: TD(0), SARSA, Q-Learning

Update after every step by leaning on the estimate you already have: when that is safe, and what it costs



Zhigang Feng

Monte Carlo Waits for the Whole Path. Temporal Difference Does Not



- ▶ Deck 3.3 averaged whole returns, $T = 300$: each sample waits T periods and carries every shock on the way.
- ▶ **This deck takes $T = 1$: one real reward, and the current estimate for everything after.**

Source: After the update-timing figure in Miao & Feng, *AI for Economists*, ch. 12, "TD(0)".

Session 3.4

Evaluation, One Step at a Time

The incremental mean, TD(0), and everything between one step and the whole path



The Sample Mean, Written One Observation at a Time

- ▶ The mean of $n + 1$ draws is the old mean plus a correction:

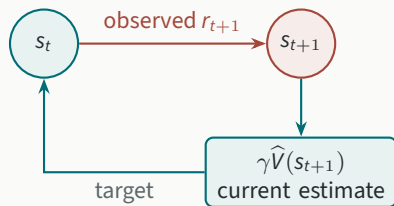
$$\bar{X}_{n+1} = \bar{X}_n + \frac{1}{n+1} (X_{n+1} - \bar{X}_n).$$

New estimate = old estimate + step size $\times$ (observation – old estimate). Deck 3.3's $\hat{Q}$ was updated exactly so.

- ▶ Replace $1/(n+1)$ by a general **step size** η_n . The **Robbins–Monro conditions** $\sum_n \eta_n = \infty, \sum_n \eta_n^2 < \infty$ give convergence: $\eta_n = n^{-\omega}$ with $\frac{1}{2} < \omega \leq 1$ qualifies.
- ▶ A constant η fails the second condition: the estimate hovers around the mean, with variance $\eta\sigma^2/(2-\eta)$.
- ▶ **A constant step forgets old data at rate $1 - \eta$. That is a virtue when the target itself moves.**

Source: H. Robbins, S. Monro, *Annals of Mathematical Statistics* 22(3), 1951; Miao & Feng, *AI for Economists*, ch. 12.

TD(0): One Real Reward, the Rest on Credit



- After each transition $(s_t, a_t, r_{t+1}, s_{t+1})$:

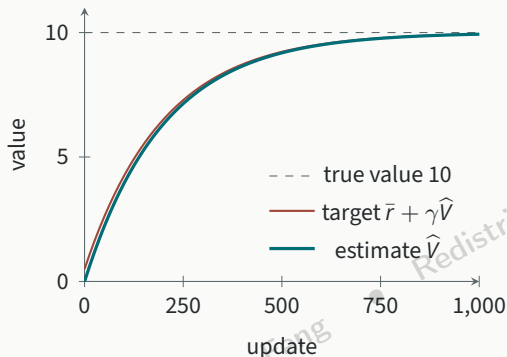
$$\hat{V}(s_t) \leftarrow \hat{V}(s_t) + \eta \delta_t,$$

$$\delta_t = r_{t+1} + \gamma \hat{V}(s_{t+1}) - \hat{V}(s_t).$$

- $r_{t+1} + \gamma \hat{V}(s_{t+1})$: the **TD target**. δ_t : the **TD error**.
- **Bootstrapping**: the estimate stands in for the unseen future.
- **Against DP: no sum over z' . Against Monte Carlo: no waiting.**

Source: R. Sutton, "Learning to predict by the methods of temporal differences", *Machine Learning* 3, 1988; Miao & Feng, ch. 12.

A Biased Target That Still Converges



- ▶ One state, reward 0.5 every period, $\gamma = 0.95$: true value 10. Start at $\hat{V} = 0$, $\eta = 0.1$.
- ▶ First three estimates: 0.050, 0.100, 0.149. The first target is only 0.5, but it is *above* the estimate.
- ▶ The expected target is $T^g \hat{V}$, a γ -contraction: always nearer the truth than $\hat{V}$ is.
- ▶ **Estimate and target rise together; the gap shrinks by $1 - \eta(1 - \gamma) = 0.995$ per update.**

Source: Arithmetic, after the worked example in Miao & Feng, *AI for Economists*, ch. 12, “Why a biased target still converges”.

TD(0) Policy Evaluation

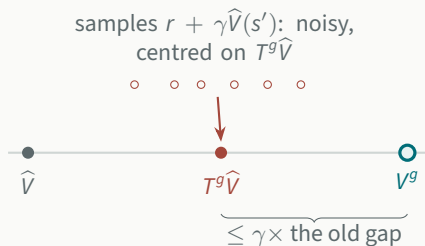
TD(0): evaluate a given policy g along one simulated path

- 1 initialise $\hat{V}(s)$; visit counts $C(s) \leftarrow 0$; $s \leftarrow s_0$
- 2 **for** $t = 0, 1, 2, \dots$ up to the step budget:
- 3 draw $a \sim g(\cdot | s)$; the simulator returns r and s'
- 4 $C(s) \leftarrow C(s) + 1$
- 5 $\delta \leftarrow r + \gamma \hat{V}(s') - \hat{V}(s)$ keep the bootstrap term even where a path is cut
- 6 $\hat{V}(s) \leftarrow \hat{V}(s) + \eta \delta$ η constant, or $C(s)^{-\omega}$
- 7 $s \leftarrow s'$

- ▶ Theorem: in a finite MDP, if every state is visited infinitely often and the step sizes satisfy Robbins–Monro, $\hat{V} \rightarrow V^g$ with probability one.
- ▶ The theorem does not say which step size is fast. $1/C(s)$ qualifies, and is unusably slow.
- ▶ **No episodes are needed: the path just runs. Cutting it is not ending it.**

Source: J. Tsitsiklis, *Machine Learning* 16, 1994; Jaakkola, Jordan & Singh, *Neural Computation* 6(6), 1994; Miao & Feng, ch. 12.

TD(0) with Indicator Features: a Table Is the Special Case



- ▶ A table is a linear critic: let $\phi(s)$ be the indicators $\mathbf{1}\{s = s_j\}$, so that ψ_j is the cell for s_j .
- ▶ Three things make it work: T^g contracts; samples centre on $T^g \hat{V}$; $C(s)^{-0.6}$ meets Robbins–Monro and restarts keep every state visited.
- ▶ Indicators share nothing, so the best fit is the target. Coarser features (Deck 3.5) give that up.
- ▶ **Measured below, fixed rule, 7×2 grid:**
0.044 at 10^5 steps, 0.028 from a 4.27 start.

Source: D. Bertsekas, J. Tsitsiklis, *Neuro-Dynamic Programming*, 1996, ch. 4; errors recorded by the book's script for ch. 12, 7×2 grid.

n -Step Returns: Between One Step and the Whole Path



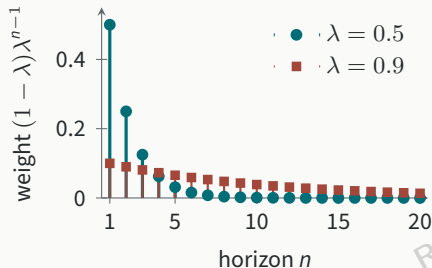
- **Deck 3.2 asked how far to roll out.** Its answer was flat: there, only $n = 1$ learned.
- The **n -step return**: n real rewards, then the estimate,

$$\hat{G}_{t:t+n} = \sum_{j < n} \gamma^j r_{t+j+1} + \gamma^n \hat{V}(s_{t+n}).$$

- $n = 1$ is TD(0); large n is Monte Carlo. Larger n : the weight γ^n falls — less bias, more variance.
- **And the update must wait n periods.**

Source: Sutton & Barto, *Reinforcement Learning*, 2018, ch. 7; Miao & Feng, *AI for Economists*, ch. 12, “ n -step returns and TD(λ)”.

The λ -Return, and the Trace That Computes It Backwards



- Deck 3.2 had to pick one n ; average every horizon at once instead:

$$\hat{G}_t^\lambda = (1 - \lambda) \sum_{n \geq 1} \lambda^{n-1} \hat{G}_{t:t+n}.$$

$\lambda = 0$: TD(0). $\lambda \rightarrow 1$: Monte Carlo.

- That needs the whole future, so take the same weights from the past:

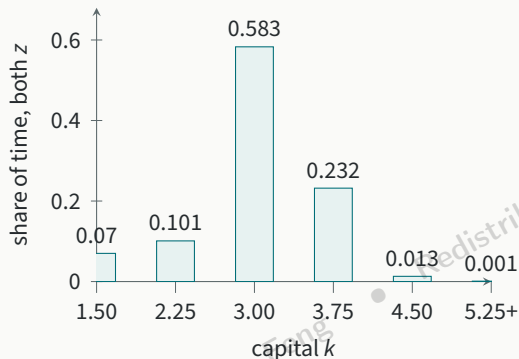
$$e_t(s) = \gamma \lambda e_{t-1}(s) + \mathbf{1}\{s_t = s\},$$

moving every state by $\eta \delta_t e_t(s)$.

- **TD(λ): one dial, and today's surprise charged to the states that led here.**

Source: Sutton & Barto, *Reinforcement Learning*, 2018, ch. 12; Singh & Sutton, *Machine Learning* 22, 1996; Miao & Feng, ch. 12.

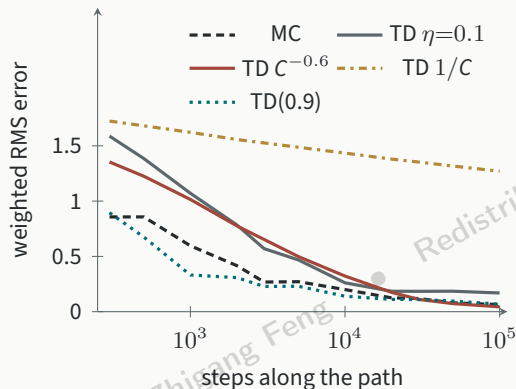
The Experiment: One Path, Five Estimators



- ▶ Evaluate g_ϵ : the optimal rule, with a uniformly random feasible action 10% of the time. Its exact values run from -0.32 to 4.56 .
- ▶ One path of 100,000 steps from $(3.75, 0.9)$; every method sees only that path. Twenty paths; medians.
- ▶ Error: root mean square, weighted by where g_ϵ spends its time.
- ▶ **Under g_ϵ , 58% of the time is spent at $k = 3.00$. The economy visits few states.**

Source: Book ch. 12 script, 7×2 grid; $\alpha = 0.36$, $\beta = 0.95$, $\delta = 0.10$, $u = \ln c$, $z \in \{0.9, 1.1\}$ with switching probability 0.2.

TD against Monte Carlo, Measured



- ▶ At 1,000 steps Monte Carlo leads (0.596 against 1.071): its target contains no estimate.
- ▶ At 100,000 steps TD(0) with $\eta = C(s)^{-0.6}$ leads: 0.044 against 0.064.
- ▶ TD(0.9) cuts the early error to a third of TD(0)'s, 0.332. The edge is gone by 100,000.
- ▶ **Bias dominates early, variance late.**

Source: Recorded by the book's script for ch. 12; initial value 0, medians of 20 paths; 7×2 grid, $\beta = 0.95$.

Target and Step Size Cannot Be Judged Apart

steps	initial value 0			initial value 4.27		
	1,000	10,000	100,000	1,000	10,000	100,000
Monte Carlo, $\eta = 1/C(s)$	0.596	0.200	0.064	0.586	0.183	0.060
TD(0), $\eta = 0.1$	1.071	0.261	0.169	1.394	0.142	0.158
TD(0), $\eta = C(s)^{-0.6}$	1.015	0.322	0.044	1.025	0.245	0.028
TD(0), $\eta = 1/C(s)$	1.622	1.434	1.270	1.690	1.484	1.319
TD(0.9), $\eta = C(s)^{-0.6}$	0.332	0.140	0.068	0.293	0.132	0.066

- ▶ $1/C(s)$ suits Monte Carlo's unbiased targets; for TD it weights early, poor targets like late, good ones. The start 4.27 is steady-state consumption valued for ever: a guess needing no data.
- ▶ Deck 3.2's $n = 5$ collapse is this one level up: its *rule* moved as the return played out, so the long target carried the rule's error, not the estimate's.
- ▶ **A sensible start and a slowly falling step size matter as much as the target.**

Source: Recorded by the book's script for ch. 12, 7×2 grid. E. Even-Dar, Y. Mansour, *JMLR* 5, 2003, on step sizes.

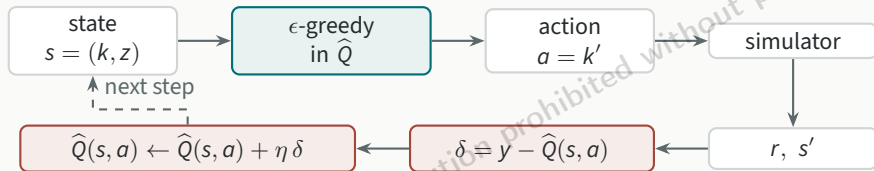
Session 3.4

Control: SARSA and Q-Learning

Improve the rule at every step, and ask whose value is being learned



From Evaluation to Control: Improve at Every Step



- Evaluation updates every step, so improvement need not wait: act on $\hat{Q}$, update it, act again.
- The object is Q again: comparing actions within a row needs no transition probabilities.
- **One choice is left: the value placed on tomorrow inside the target y .**

Source: After the control-loop figure in Miao & Feng, *AI for Economists*, ch. 12, “Temporal-difference control”.

SARSA, Derived: What an On-Policy Target Converges To

- ▶ Deck 3.2 named this road and had no table to take it: there the state and the action are both real numbers. The 7×2 grid has cells. The target values the action that will really be taken,

$$y_{t+1} = r_{t+1} + \gamma \hat{Q}(s_{t+1}, a_{t+1}), \quad a_{t+1} \sim g_\epsilon(\cdot | s_{t+1}).$$

- ▶ Average over that draw: the expected target is $T^{g_\epsilon} \hat{Q}$, whose fixed point is Q^{g_ϵ} . **SARSA** converges to the value of the rule that generated the data — **on-policy**.
- ▶ With ϵ held at 0.1 that rule is not g^* but the best **ϵ -soft policy**: on the 7×2 grid that limit alone costs 0.18%, before any learning error.
- ▶ So the defect is the estimand, not the data. On the lab's 20×2 grid the gap survives: 0.69–1.47% against Q-learning's 0.36–0.39%, three seeds.
- ▶ **To reach g^* , ϵ must fall to zero slowly: greedy in the limit, with infinite exploration (GLIE).**

Source: Rummery & Niranjan, 1994; Singh et al., *Machine Learning* 38, 2000; 20×2 numbers from the book's script, ch. 12.

Q-Learning, Derived: What the Maximum Buys, and What It Costs

- ▶ One character separates it from SARSA: the next action is maximised, over the feasible set alone,

$$y_{t+1} = r_{t+1} + \gamma \max_{a' \in \mathcal{A}(s_{t+1})} \hat{Q}(s_{t+1}, a').$$

- ▶ Its expected target is $T^* \hat{Q}$, a γ -contraction with fixed point Q^* : greedy **target policy**, ϵ -greedy **behaviour policy**, hence **off-policy**. A fixed $\epsilon > 0$ costs nothing, given bounded rewards, Robbins–Monro steps, and every feasible pair updated for ever.
- ▶ So the defect is the data, not the estimand: where the path rarely goes a cell stays wrong — 4 states of 14 at 10^6 steps.
- ▶ **Maximisation bias**: $E[\max_{a'} \hat{Q}] \geq \max_{a'} E[\hat{Q}]$, so noise lifts the target and bootstrapping passes it on. Large when the actions are many.
- ▶ **Off-policy buys the right target, not the right data: the best portfolio's value, learned while holding a cautious one.**

Source: C. Watkins, P. Dayan, “Q-learning”, *Machine Learning* 8, 1992; Miao & Feng, *AI for Economists*, ch. 12.

Why Q-Learning Needs No Importance Ratio

- ▶ Deck 3.3: data from one rule, used to value another, must be reweighted by g/b at every period of the return.
- ▶ One-step Q-learning uses a single transition. Given (s_t, a_t) , the next state comes from $P(\cdot \mid s_t, a_t)$: the behaviour rule plays no part.
- ▶ The $\max$ draws no action from the behaviour rule at all.
- ▶ The only action it drew is a_t , and the update is *for* the cell (s_t, a_t) . Conditional on it, no ratio appears.
- ▶ **The behaviour rule only decides which cells get data. That still matters, as the measurements will show.**

Source: Miao & Feng, *AI for Economists*, ch. 12, “Q-learning”; Watkins & Dayan, *Machine Learning* 8, 1992.

One Algorithm, One Line of Difference

TD control: SARSA and Q-learning

- 1 initialise $\hat{Q}(s, a)$ on feasible cells; counts $C(s, a) \leftarrow 0$; $s \leftarrow s_0$
- 2 choose a in $\mathcal{A}(s)$, ϵ -greedy in $\hat{Q}$
- 3 **for** $t = 0, 1, 2, \dots$ up to the step budget:
- 4 take a ; the simulator returns r and s'
- 5 choose a' in $\mathcal{A}(s')$, ϵ -greedy in $\hat{Q}$
- 6 **SARSA:** $y \leftarrow r + \gamma \hat{Q}(s', a')$ **Q-learning:** $y \leftarrow r + \gamma \max_{a'' \in \mathcal{A}(s')} \hat{Q}(s', a'')$
- 7 $C(s, a) \leftarrow C(s, a) + 1$; $\hat{Q}(s, a) \leftarrow \hat{Q}(s, a) + C(s, a)^{-0.6} [y - \hat{Q}(s, a)]$; $(s, a) \leftarrow (s', a')$

Line 6 is the whole difference. Deploy $g(s) = \arg \max_{a \in \mathcal{A}(s)} \hat{Q}(s, a)$ in both: **exploration is for learning, not for use.**

Source: Miao & Feng, *AI for Economists*, ch. 12, “Q-learning” (algorithm for TD control). This is the lab’s block C.

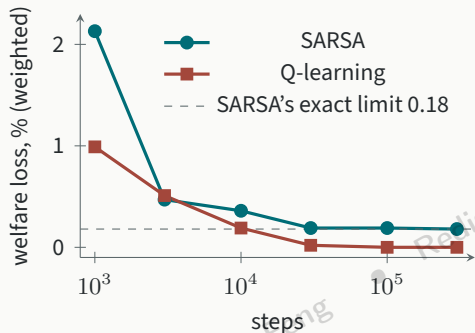
What Each Converges To: Q^* against Q_ϵ

next capital k' , from state $(k, z) = (2.25, 1.1)$	1.50	2.25	3.00
Q^* : what Q-learning approaches	1.799	2.524	2.546
Q_ϵ : what SARSA approaches, $\epsilon = 0.1$	1.000	1.676	1.605

- ▶ Both limits can be computed exactly. Their greedy choices differ in 2 of 14 states, and the same way: SARSA's limit saves one grid step less.
- ▶ Raising k' from 2.25 to 3.00 costs 0.919 of utility today. The discounted gain tomorrow is 0.941 under g^* , worth it; only 0.848 under the ϵ -soft rule, not worth it.
- ▶ Every future period carries a 10% chance of a random k' , so part of today's extra capital will be spent at random. Exploration noise dilutes the return to saving, at every grid step.
- ▶ **At $\epsilon = 0.3$ the rules differ in 4 states. All four save less.**

Source: Book ch. 12 script, 7×2 grid; $\alpha = 0.36, \beta = 0.95, \delta = 0.10, u = \ln c, z \in \{0.9, 1.1\}$ with switching probability 0.2.

Measured: 300,000 Steps, Twenty Runs



at 300,000 steps, medians	SARSA	Q
loss, weighted	0.18%	0.00%
loss, plain mean, 14 states	1.66%	2.01%
states with a wrong action	4	4
cells with 10 or more samples	43	42

- By 10,000 steps: 0.36% and 0.19%, weighted; Q-learning's last error goes by 100,000. Its 0.00% is the row max Deck 3.2 had no table for.
- **The 2% plain mean comes from $k \geq 4.50$: under 1.5% of the time.**

Source: Recorded by the book's script for ch. 12, 7×2 grid, $\epsilon = 0.1$, 20 runs. Loss: consumption-equivalent, against g^* .

Session 3.4

Using Samples Well

Exploration pace, lower-variance targets, and where TD sits among all backups



Sampling and Step Sizes: Five Fixes, One Idea

- ▶ **Pace.** A fixed ϵ parks SARSA at the ϵ -soft optimum; $\epsilon = 1/\sqrt{C(s)}$ satisfies GLIE. Boltzmann (Deck 3.3) wastes fewer trials on plainly bad actions.
- ▶ **Start. Optimistic initial values:** 4.27 exceeds most true values, so untried actions look good and get tried.
- ▶ **Step.** $\eta = 1/C$ can need steps growing exponentially in $1/(1 - \gamma)$; $C^{-\omega}$, $\frac{1}{2} < \omega < 1$, polynomially. The lab uses $\omega = 0.6$.
- ▶ **Target. Expected SARSA** replaces the drawn a_{t+1} by $\sum_{a'} g_{\epsilon}(a' | s_{t+1}) \hat{Q}(s_{t+1}, a')$: same expectation, less variance. The greedy rule in that sum is Q-learning.
- ▶ **Old data.** Multi-step off-policy targets need importance ratios, or a cut at the first non-greedy action; from history, the behaviour rule must itself be estimated.
- ▶ **Two estimators.** One table picks the action, an independent one values it: the max bias goes. **Every fix is where samples come from, or how a target uses them.**

Source: Sutton & Barto, 2018, §6.6; Even-Dar & Mansour, *JMLR* 5, 2003; H. van Hasselt, *NeurIPS*, 2010; Imbens & Rubin, 2015.

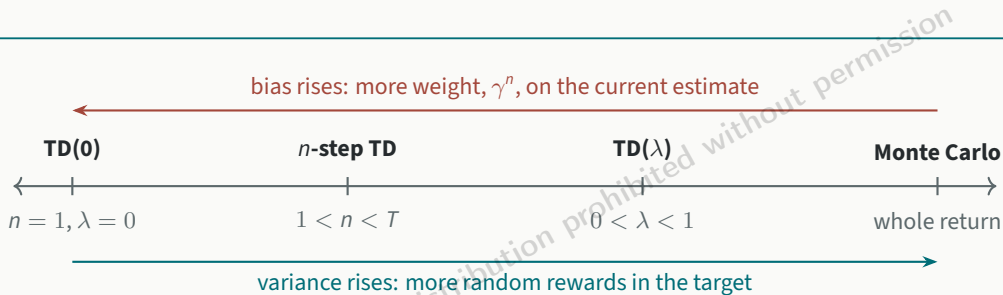
The Backup Spectrum: Two Choices Define an Evaluation Method

real rewards	full expectation over s' (needs P)	one sampled s' (no P)
one period	one DP backup: a sweep of value iteration	TD(0) : target $r + \gamma \hat{V}(s')$
n periods	modified policy iteration: roll the rule forward n periods	n-step TD, TD(λ)
the whole path	exact evaluation: solve the linear system	Monte Carlo : the return G_t alone

- ▶ Columns: is P known? Rows: how much the target leans on the current estimate.
- ▶ **Decks 3.1 to 3.4 filled this table. Every cell is the same Bellman equation.**

Source: Miao & Feng, *AI for Economists*, ch. 12, “One mathematical skeleton, different solution paths” (table).

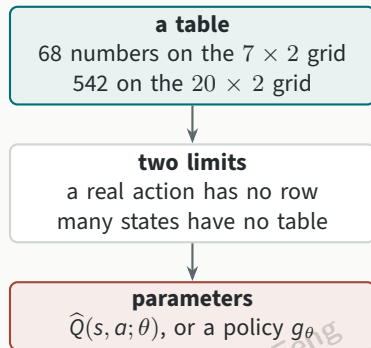
One Dial: Bias against Variance



- ▶ TD(0): one random reward, low variance; a poor estimate leaks into every target.
- ▶ Monte Carlo: no estimate in the target, so no such bias; every shock along the path is in it.
- ▶ **Deck 3.2's actor-critic sat at the TD end from its first line: an update every step, driving the rule as it goes.**

Source: Miao & Feng, *AI for Economists*, ch. 12, "One mathematical skeleton"; Sutton & Barto, 2018, ch. 7 and 12.

What a Q Table Cannot Do



- ▶ Done so far: the expectation is sampled (3.3), the update is one step (3.4), evaluation and improvement interleave.
- ▶ **Two limits, not one.** A real action has no row to max over; many states have no table to hold — challenge 1 of Deck 3.1, untouched here.
- ▶ Deck 3.2 stepped past both: a gradient instead of a max, five weights instead of cells. But every theorem here assumes the table.
- ▶ **Deck 3.5 says when that is safe — linear features, then a network — and why a nudge of $\delta \nabla \log g_\theta$ points uphill.**

Source: Cell counts: Session 3 lab, block A. Miao & Feng, *AI for Economists*, ch. 12, “Function approximation”.